Company:		Page:	1
Address:		Specifier:	
Phone I Fax:	|	E-Mail:	
Design:	2 bolt- Concrete - Jul 8, 2022	Date:	08/07/2022
Fastening point:			

Specifier's comments:

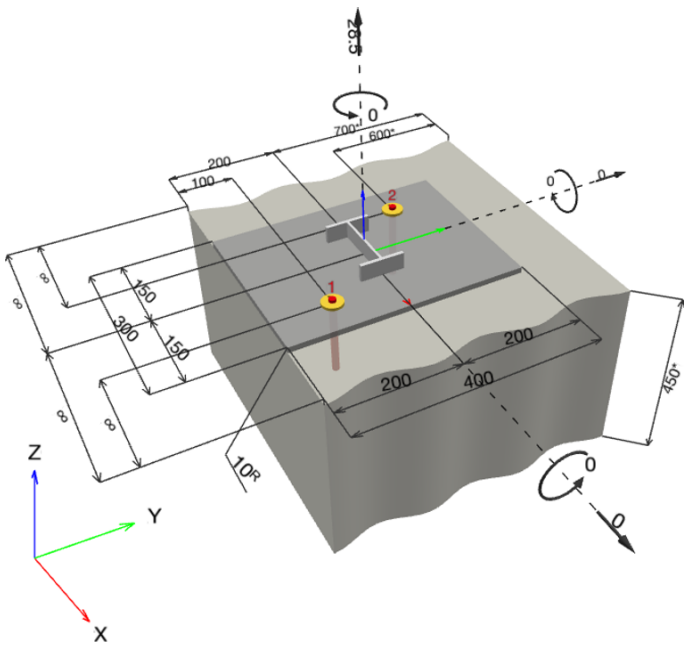
1 Input data



Anchor type and diameter:	**HIT-RE 500 V4 + HAS-U 5.8 M12**
Return period (service life in years):	50
Item number:	2223823 HAS-U 5.8 M12x160 (element) / not available (adhesive)
Filling set or any suitable annular gap filling solution	
Effective embedment depth:	$h_{ef,act}$ = 120.0 mm ($h_{ef,limit}$ = - mm)
Material:	5.8
Evaluation Service Report:	Hilti Technical Data
Issued I Valid:	- | -
Proof:	Engineering judgement SOFA BOND - based on ETAG BOND testing
Stand-off installation:	e_b = 0.0 mm (no stand-off); t = 10.0 mm
Anchor plate[R]:	l_x x l_y x t = 300.0 mm x 400.0 mm x 10.0 mm; (Recommended plate thickness: not calculated)
Profile:	ISLB, ISLB 125; (L x W x T x FT) = 125.0 mm x 75.0 mm x 4.4 mm x 6.5 mm
Base material:	cracked concrete, M 20, $f_{c,cube}$ = 20.00 N/mm^2; h = 450.0 mm, Temp. short/long: 40/24 °C
Installation:	**hammer drilled hole, Installation condition: Dry**
Reinforcement:	no reinforcement or reinforcement spacing >= 150 mm (any Ø) or >= 100 mm (Ø <= 10 mm) no longitudinal edge reinforcement Reinforcement to control splitting according to EOTA TR 029, 5.2.2.6 present.

[R] - The anchor calculation is based on a rigid anchor plate assumption.

Geometry [mm] & Loading [kN, kNm]

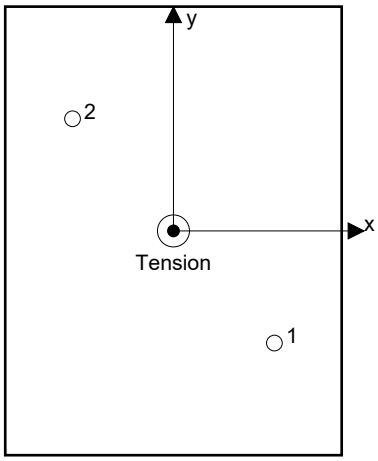


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1.1 Load combination

Case	Description	Forces [kN] / Moments [kNm]	Seismic	Fire	Max. Util. Anchor [%]
1	Combination 1	N = 28.500; V_x = 0.000; V_y = 0.000; M_x = 0.000; M_y = 0.000; M_z = 0.000;	no	no	75

2 Load case/Resulting anchor forces



Anchor reactions [kN]
Tension force: (+Tension, -Compression)

Anchor	Tension force	Shear force	Shear force x	Shear force y
1	14.250	0.000	0.000	0.000
2	14.250	0.000	0.000	0.000

max. concrete compressive strain: - [‰]
max. concrete compressive stress: - [N/mm^2]
resulting tension force in (x/y)=(0.0/0.0): 28.500 [kN]
resulting compression force in (x/y)=(0.0/0.0): 0.000 [kN]

Anchor forces are calculated based on the assumption of a rigid anchor plate.

Input data and results must be checked for conformity with the existing conditions and for plausibility!
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3 Tension load (EOTA TR 029, Section 5.2.2)

	Load [kN]	Capacity [kN]	Utilization β_N [%]	Status
Steel Strength*	14.250	28.133	51	OK
Combined pullout-concrete cone failure**	28.500	40.003	72	OK
Concrete Breakout Failure**	28.500	38.042	75	OK
Splitting failure**	N/A	N/A	N/A	N/A

* highest loaded anchor **anchor group (anchors in tension)

3.1 Steel Strength

$N_{Rk,s}$ [kN]	$\gamma_{M,s}$	$N_{Rd,s}$ [kN]	N_{Sd} [kN]
42.200	1.500	28.133	14.250

3.2 Combined pullout-concrete cone failure

$A_{p,N}$ [mm²]	$A^0_{p,N}$ [mm²]	$\tau_{Rk,ucr,25}$ [N/mm²]	$s_{cr,Np}$ [mm]	$c_{cr,Np}$ [mm]	c_{min} [mm]
201,600	129,600	18.00	360.0	180.0	100.0

ψ_c	$\tau_{Rk,cr}$ [N/mm²]	k	$\psi^0_{g,Np}$	$\psi_{g,Np}$
0.894	9.84	2.300	1.000	1.000

$e_{c1,N}$ [mm]	$\psi_{ec1,Np}$	$e_{c2,N}$ [mm]	$\psi_{ec2,Np}$	$\psi_{s,Np}$	$\psi_{re,Np}$
0.0	1.000	0.0	1.000	0.867	1.000

$N^0_{Rk,p}$ [kN]	$N_{Rk,p}$ [kN]	$\gamma_{M,p}$	$N_{Rd,p}$ [kN]	N_{Sd} [kN]
44.509	60.005	1.500	40.003	28.500

Group anchor ID
1, 2

3.3 Concrete Breakout Failure

$A_{c,N}$ [mm²]	$A^0_{c,N}$ [mm²]	$c_{cr,N}$ [mm]	$s_{cr,N}$ [mm]
201,600	129,600	180.0	360.0

$e_{c1,N}$ [mm]	$\psi_{ec1,N}$	$e_{c2,N}$ [mm]	$\psi_{ec2,N}$	$\psi_{s,N}$	$\psi_{re,N}$
0.0	1.000	0.0	1.000	0.867	1.000

k_1	$N^0_{Rk,c}$ [kN]	$\gamma_{M,c}$	$N_{Rd,c}$ [kN]	N_{Sd} [kN]
7.200	42.327	1.500	38.042	28.500

Group anchor ID
1, 2

Input data and results must be checked for conformity with the existing conditions and for plausibility!
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4 Shear load (EOTA TR 029, Section 5.2.3)

	Load [kN]	Capacity [kN]	Utilization β_V [%]	Status
Steel Strength (without lever arm)*	N/A	N/A	N/A	N/A
Steel failure (with lever arm)*	N/A	N/A	N/A	N/A
Pryout Strength*	N/A	N/A	N/A	N/A
Concrete edge failure in direction **	N/A	N/A	N/A	N/A

* highest loaded anchor **anchor group (relevant anchors)

5 Displacements (highest loaded anchor)

Short term loading:

N_{Sk} = 10.556 [kN] δ_N = 0.1167 [mm]

V_{Sk} = 0.000 [kN] δ_V = 0.0000 [mm]

δ_{NV} = 0.1167 [mm]

Long term loading:

N_{Sk} = 10.556 [kN] δ_N = 0.3267 [mm]

V_{Sk} = 0.000 [kN] δ_V = 0.0000 [mm]

δ_{NV} = 0.3267 [mm]

Comments: Tension displacements are valid with half of the required installation torque moment for uncracked concrete! Shear displacements are valid without friction between the concrete and the anchor plate! The gap due to the drilled hole and clearance hole tolerances are not included in this calculation!

The acceptable anchor displacements depend on the fastened construction and must be defined by the designer!

6 Warnings

- The anchor design methods in PROFIS Engineering require rigid anchor plates per current regulations (AS 5216:2021, ETAG 001/Annex C, EOTA TR029 etc.). This means load re-distribution on the anchors due to elastic deformations of the anchor plate are not considered - the anchor plate is assumed to be sufficiently stiff, in order not to be deformed when subjected to the design loading. PROFIS Engineering calculates the minimum required anchor plate thickness with CBFEM to limit the stress of the anchor plate based on the assumptions explained above. The proof if the rigid anchor plate assumption is valid is not carried out by PROFIS Engineering. Input data and results must be checked for agreement with the existing conditions and for plausibility!
- Checking the transfer of loads into the base material is required in accordance with EOTA TR 029, Section 7!
- The design is only valid if the clearance hole in the fixture is not larger than the value given in Table 4.1 of EOTA TR029! For larger diameters of the clearance hole see Chapter 1.1. of EOTA TR029!
- The accessory list in this report is for the information of the user only. In any case, the instructions for use provided with the product have to be followed to ensure a proper installation.
- Characteristic bond resistances depend on short- and long-term temperatures.
- The design method SOFA assumes that no hole clearance between the anchors and the fixture is present. This can be achieved by filling the gap with mortar of sufficient compressive strength (e.g. by using the Hilti Filling set) or by other suitable means
- The compliance with current standards (e.g. EN 1993, AS 4100:1998, etc.) is the responsibility of the user
- An SLS-check is not performed for SOFA and has to be provided by the user!
- The characteristic bond resistances depend on the return period (service life in years): 50

Fastening meets the design criteria!

Input data and results must be checked for conformity with the existing conditions and for plausibility!
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7 Installation data

Anchor plate, steel: E 250 (Fe 410 W) B; E = 200,000.00 N/mm^2; f_{yk} = 250.00 N/mm^2

Profile: ISLB, ISLB 125; (L x W x T x FT) = 125.0 mm x 75.0 mm x 4.4 mm x 6.5 mm

Hole diameter in the fixture: d_f = 14.0 mm

Plate thickness (input): 10.0 mm

Recommended plate thickness: not calculated

Drilling method: Hammer drilled

Cleaning: Compressed air cleaning of the drilled hole according to instructions for use is required

Anchor type and diameter: HIT-RE 500 V4 + HAS-U 5.8 M12

Item number: 2223823 HAS-U 5.8 M12x160 (element) / not available (adhesive)

Maximum installation torque: 40 Nm

Hole diameter in the base material: 14.0 mm

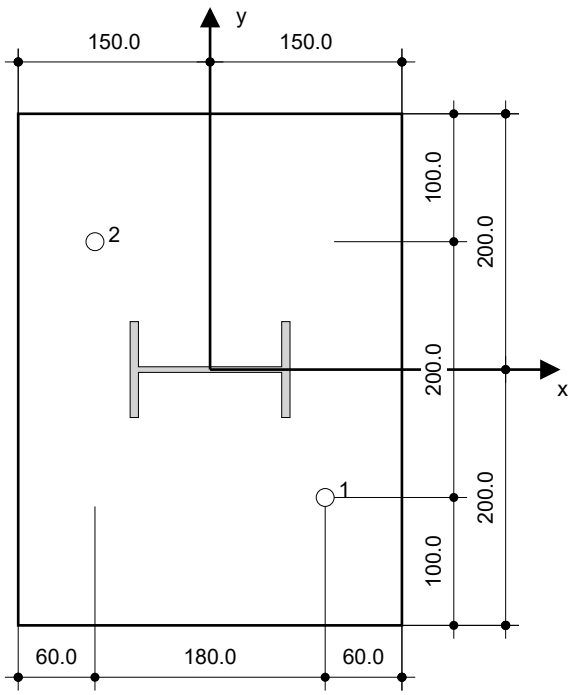
Hole depth in the base material: 120.0 mm

Minimum thickness of the base material: 150.0 mm

Hilti HAS-U threaded rod with HIT-RE 500 V4 injection mortar with 120 mm embedment h_ef, M12, Steel galvanized, Hammer drilled installation per instruction for use, with annular gaps filled with Hilti Filling set or any suitable gap solutions

7.1 Recommended accessories

Drilling	Cleaning	Setting
• Suitable Rotary Hammer • Properly sized drill bit	• Compressed air with required accessories to blow from the bottom of the hole • Proper diameter wire brush	• Dispenser including cassette and mixer • Filling set • Torque wrench



Coordinates Anchor [mm]

Anchor	x	y	c_{-x}	c_{+x}	c_{-y}	c_{+y}
1	90.0	-100.0	-	-	100.0	800.0
2	-90.0	100.0	-	-	300.0	600.0

Input data and results must be checked for conformity with the existing conditions and for plausibility!
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